



Mini Project Report On

ReCo: Retail Commerce using Image Recognition and ML

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award of the degree of*

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By

ABHISHIKTH S MATTOM (U2303011)

ALAN JOPHY (U2303022)

ARNOLD SHIBU (U2303052)

ALEN ABRAHAM SAJI (U2303027)

Under the guidance of

Dr. AMITHA MATHEW

**Department of Computer Science & Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Affiliated to APJ Abdul Kalam Technological University)**

Rajagiri Valley, Kakkanad, Kochi, 682039

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CERTIFICATE

*This is to certify that the mini project report entitled “**ReCo: Retail Commerce using Image Recognition and ML**” is a bonafide record of the work done by **ALAN JOPHY (U2303022)**, **ABHISHIKTH S MATTOM (U2303011)**, **ARNOLD SHIBU (U2303052)**, **ALEN ABRAHAM SAJI (U2303027)**, submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025–2026.*

Project Guide
Dr. Amitha Mathew
Assistant Professor
Dept. of CSE
RSET

Project Coordinator
Dr. Mary Priya Sebastian
Associate Professor
Dept. of CSE
RSET

Dr. Preetha K.G.
Professor, Head of the Department
Dept. of CSE
RSET

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ABHISHIKTH S MATTOM

ALAN JOPHY

ARNOLD SHIBU

ALEN ABRAHAM SAJI

Abstract

ReCo is an advanced retail automation system designed to streamline the checkout process in FMCG stores through real-time product identification. The system leverages a live camera feed processed by an EfficientNet-B0 deep learning model to automatically recognize consumer goods at the point of sale. To ensure operational reliability under challenging retail conditions—such as poor lighting or visual occlusion—the architecture includes an automatic barcode-scanning fallback mechanism and a manual entry option. The ecosystem features two primary interfaces: a mobile application for shopkeepers that facilitates rapid billing, dynamic payment QR code generation, and local inventory tracking, and a web-based dashboard for manufacturers to monitor product movement. By integrating a cloud-based MySQL database, the system utilizes machine learning to analyze aggregated sales data, offering actionable insights into demand forecasting, market trends, and inventory optimization for both shopkeepers and FMCG manufacturers.

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List of Abbreviations

ML	-	Machine Learning
FMCG	-	Fast-Moving Consumer Goods
QR	-	Quick Response
UPI	-	Unified Payments Interface
CNN	-	Convolutional Neural Network
SRS	-	Software Requirements Specification
ARIMA	-	AutoRegressive Integrated Moving Average
UI	-	User Interface
API	-	Application Programming Interface
POS	-	Point of Sale
SQL	-	Structured Query Language

Chapter 1

Introduction

1.1 Background

Retail commerce generates a large volume of transactional data every day, especially in small and medium retail stores. However, many small retailers still rely on manual billing systems or basic digital tools that lack advanced analytical capabilities. These traditional approaches often lead to inefficiencies such as slower checkout processes, human errors in billing, and limited visibility into sales trends and inventory performance.

Recent advancements in computer vision and machine learning have enabled automated systems that can recognize objects and extract meaningful insights from data. In the retail sector, image recognition techniques can be used to identify products from their packaging, enabling automated checkout processes and reducing the need for manual product entry or specialized scanning hardware. At the same time, machine learning models can analyze historical sales data to generate valuable insights such as product performance trends, demand forecasting, and inventory optimization.

The growing availability of mobile devices with powerful cameras and processing capabilities makes it possible to implement such intelligent retail systems using mobile applications. By combining image recognition for product identification, barcode scanning as a fallback mechanism, and machine learning techniques for sales analysis, retailers can automate checkout operations while gaining useful business insights. This approach can significantly improve operational efficiency, reduce hardware costs, and support data-driven decision making for both shopkeepers and manufacturers.

1.2 Problem Definition

Small and medium retail stores commonly rely on manual billing or basic point-of-sale systems that require dedicated hardware such as barcode scanners and billing terminals.

These traditional methods can slow down the checkout process, increase the chances of human errors in product entry, and make it difficult for retailers to efficiently track and manage sales records.

Additionally, many small retailers lack effective tools to analyze their sales data, making it challenging to identify product demand patterns, monitor inventory levels, and understand sales performance over time. The absence of automated data collection and analysis limits their ability to make informed business decisions and respond quickly to changing market demands.

1.3 Scope and Motivation

The scope of this project is to develop a smart retail commerce system that automates product identification and sales tracking in retail stores. The system uses image recognition to identify products during checkout and barcode scanning as a fallback mechanism when recognition fails. It records sales transactions automatically and stores them for further analysis and demand forecasting.

The motivation for this project comes from the challenges faced by small retail stores that rely on manual billing or basic point-of-sale systems. These methods are often slow, prone to human errors, and provide limited insights into sales performance. By introducing an intelligent system that combines image recognition and machine learning-based analytics, the project aims to improve checkout efficiency, reduce hardware dependency, and help retailers make better data-driven decisions.

1.4 Objectives

- Develop a real-time product recognition pipeline using the EfficientNet-B0 model.
- Implement a rapid automatic barcode-scanning fallback mechanism.
- Design an intuitive shopkeeper billing interface.
- Provide predictive analytics for demand forecasting.
- Develop a web dashboard for manufacturers.

1.5 Challenges

- Processing image recognition algorithms in real-time requires significant computational efficiency to minimize latency on standard mobile devices.
- Accurately identifying products under varying lighting conditions, glare, and physical occlusions common in busy retail environments.
- Continuously updating the model to recognize an ever-expanding inventory of FMCG products that often feature highly similar packaging.

1.6 Assumptions

- The retail store has stable internet connectivity to sync with the cloud database .
- Shopkeepers utilize a smartphone or tablet equipped with a functional camera capable of clear image capture.
- Standard barcode labels remain available on FMCG products to serve as an effective fallback identification method.

1.7 Societal / Industrial Relevance

Reco bridges the digital divide for small-to-medium retail vendors by replacing traditional, error-prone manual bookkeeping with a streamlined digital ecosystem, equipping them with enterprise-grade automation without the need for expensive point-of-sale hardware. This digital transformation empowers local shopkeepers to remain competitive against larger supermarket chains by automating inventory tracking and sales logging through an EfficientNet-B0 recognition pipeline. On an industrial level, the platform provides FMCG manufacturers with a specialized web dashboard that offers valuable, aggregated insights into consumer purchasing behavior and regional product demand. By centralizing real-time transaction data in a cloud-based MySQL database and applying ARIMA modeling for demand forecasting, ReCo optimizes supply chain logistics, reduces inventory waste, and facilitates data-driven decision-making across the entire FMCG value chain.

1.8 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 details the Software Requirements Specification (SRS), covering the overall system features and external interface requirements. Chapter 3 discusses the System Architecture and Design, including the dataset identified, proposed deep learning methodologies, and database structure. The report concludes with the system implementation modules, work schedule, and references.

Chapter 2

Software Requirements Specification

2.1 Introduction

This chapter presents the Software Requirements Specification (SRS) for the proposed Smart Retail Commerce System. The SRS document defines the functional and non-functional requirements of the system and provides a detailed description of the expected behavior, features, and operational constraints of the application. It serves as a reference for developers, evaluators, and stakeholders involved in the development and implementation of the system. The document outlines the requirements necessary for designing a system that automates retail product identification, sales tracking, and data analysis in retail environments.

The proposed system improves retail checkout efficiency by using image recognition to identify products through a mobile device camera, with barcode scanning used as a fallback mechanism when recognition fails. Once the product is identified, the system records the transaction and stores the sales data for further analysis. The collected data is then used to generate insights such as sales trends, product performance, and demand forecasting. These insights are presented through analytics dashboards for shopkeepers, while aggregated sales insights are also provided to manufacturers through a dedicated manufacturer dashboard to help understand regional demand patterns and product performance.

2.2 Overall Description

The Smart Retail Commerce System is designed to automate product identification and sales tracking in retail environments using modern machine learning technologies. The system allows shopkeepers to scan product packaging using a mobile application, where an image recognition model identifies the product in real time. If the image recognition

process fails or the confidence level is low, barcode scanning is used as a fallback mechanism to ensure accurate product identification. Once the product is identified, the system records the transaction and stores it as timestamped sales data.

The system also includes data analytics capabilities that process stored sales records to generate useful business insights. Retailers can view analytics such as sales trends, fast-moving products, and inventory requirements through a shopkeeper dashboard. Additionally, aggregated and anonymized sales insights are provided to manufacturers through a manufacturer dashboard, enabling them to understand regional demand patterns and product performance. This overall system helps improve checkout efficiency, reduce manual effort, and support data-driven decision-making for both retailers and manufacturers.

2.3 External Interface Requirements

User Interface: The system provides a mobile application interface that allows shopkeepers to scan product packaging, view billing information, and access sales analytics dashboards. Manufacturers can access aggregated sales insights through a web-based dashboard.

Hardware Interface: The application interfaces with the device camera for real-time image capture of product packaging and barcode scanning during checkout.

Software and Database Interface: The system connects to a cloud-based MySQL database through secure APIs to store product information, sales transactions, and inventory data. This enables synchronized digital bookkeeping and analytics processing for both retailers and manufacturers.

2.4 System Features

The system provides automated retail billing using computer vision-based product identification through a mobile device camera. When image recognition fails or confidence is low, barcode scanning is used as a fallback mechanism to ensure accurate product identification during checkout. Each transaction is automatically recorded and stored as timestamped sales data.

The system also performs automatic stock deduction after every sale and maintains updated inventory records. Retailers can view sales metrics, product performance, and inventory status through a dashboard interface. Additionally, machine learning techniques such as ARIMA are used to analyze historical sales data and generate demand predictions. Aggregated analytics are also presented through a manufacturer dashboard to provide insights into regional demand patterns and product performance.

2.5 Other Nonfunctional Requirements

1. Performance

The system should perform product recognition and barcode scanning with minimal delay to ensure smooth checkout operations. Efficient processing is required so that the shopkeeper can scan products quickly without slowing down the billing process.

2. Reliability

The application should operate continuously without crashes and maintain accurate transaction and inventory records. The system must ensure that sales data is stored correctly and that inventory updates are consistently reflected in the database.

3. Security

Access to sales data and analytics dashboards must be restricted to authorized users such as shopkeepers and manufacturers. Proper authentication mechanisms should be implemented to protect sensitive business information.

4. Usability

The system should provide a simple and user-friendly interface so that shopkeepers can easily perform product scanning and billing operations. The interface should be intuitive and require minimal technical knowledge for operation.

5. Scalability

The system should be able to handle increasing numbers of transactions and users without significant performance degradation. This ensures that the system remains efficient as the number of retail stores and data volume grows.

Chapter 3

System Architecture and Design

3.1 System Overview

The Smart Retail Commerce System is designed to automate product identification and retail transaction management using image recognition and machine learning techniques. The system enables shopkeepers to scan product packaging using a mobile application, where the device camera captures the product image. A trained image recognition model processes the captured image and identifies the product in real time. If the image recognition process fails or produces low confidence results, the system activates barcode scanning as a fallback mechanism to ensure accurate product identification during checkout.

After the product is successfully identified, the system automatically records the transaction and stores it as timestamped sales data in a centralized database. The system also updates inventory information by deducting the sold quantity from available stock. This digital record keeping eliminates manual billing errors and simplifies sales tracking for retail store owners.

The stored sales data is further analyzed using machine learning techniques to generate meaningful business insights. Sales trends, fast-moving products, and stock requirements can be visualized through a shopkeeper dashboard within the application. In addition to retailer insights, aggregated and anonymized sales data is provided through a manufacturer dashboard, allowing manufacturers to analyze regional demand patterns and product performance across different retail locations.

Overall, the system integrates product recognition, automated billing, inventory tracking, and analytics into a unified platform that improves checkout efficiency, reduces dependency on specialized POS hardware, and supports data-driven decision making in retail environments. The overall flow of the system is illustrated in Figure 3.1.

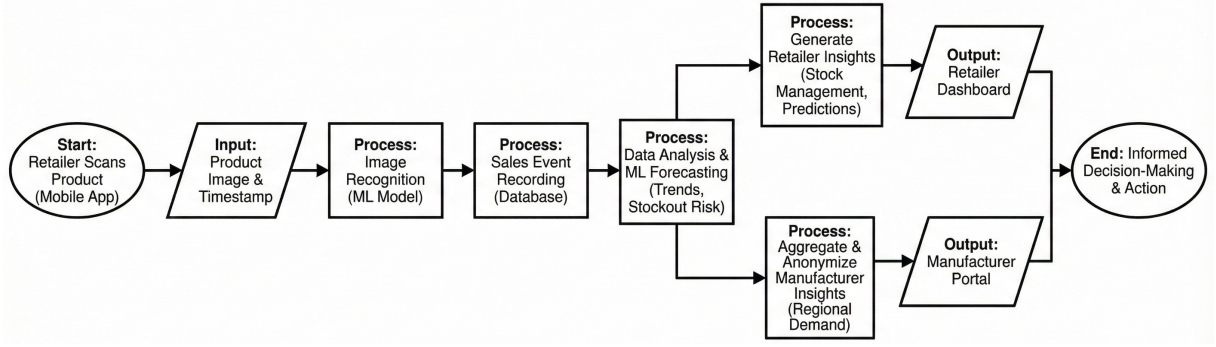


Figure 3.1: Flow Diagram

3.2 Architectural Design

The Smart Retail Commerce System follows a modular architecture that integrates computer vision, data processing, and analytics components to automate retail billing and inventory management. The system begins with a live camera feed that captures images of product packaging during checkout. These images are passed through the input layer, which forwards the captured frames to the product identification modules for further processing.

The primary identification is performed by the Image Recognition Module, which uses a trained machine learning model to classify the captured product image and generate a product identifier along with a confidence score. If the confidence level is low or the product cannot be identified accurately, the system activates the Barcode Scanning Module as a fallback mechanism. This ensures reliable product identification even in cases where image recognition may fail.

The results from these identification modules are then sent to the Data Integration and Processing Module. This component validates the product information, prepares the transaction data, and forwards it to the Billing System. The billing system records the transaction and stores it in the Cloud Database (MySQL), which maintains product details, sales records, and inventory information.

The Inventory Management Module interacts with the database to automatically update stock levels after each sale. Additionally, the stored sales data is used by the Analytics Dashboard to generate business insights such as sales trends, product performance, and demand patterns. These insights are available to retailers through a dashboard interface,

while aggregated analytics can also be accessed by manufacturers to understand regional product demand and market trends. The complete system architecture is illustrated in Figure 3.2, and the interaction between system components is shown in Figure 3.3.

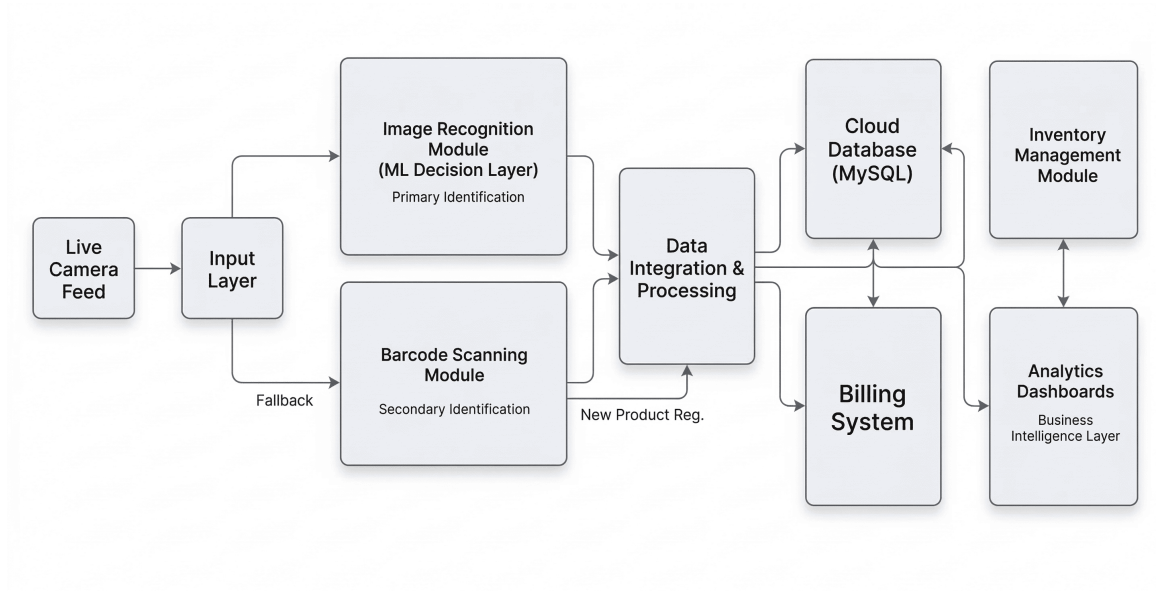


Figure 3.2: System Architecture

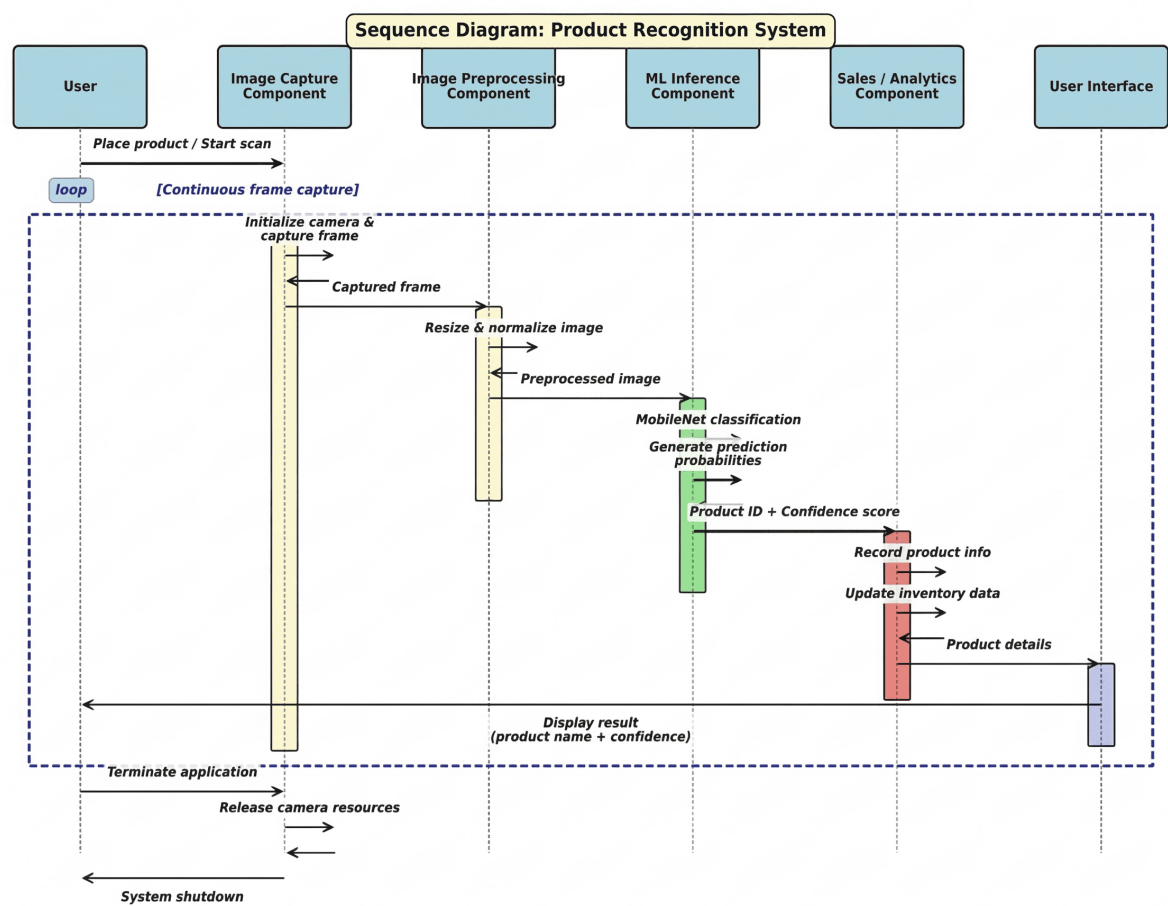


Figure 3.3: Interaction Diagram

3.3 Dataset Identified

The dataset used for training the product recognition model is the “Retail Store Indian Trial” dataset. It consists of labeled images of retail grocery products commonly found in Indian retail stores. The dataset contains images of various packaged products such as food items, household goods, and daily-use consumer products. These images represent real retail scenarios and include variations in lighting conditions, viewing angles, and backgrounds to simulate practical store environments.

Each image in the dataset is annotated with the corresponding product label, enabling supervised training of the image recognition model. The dataset contains approximately 754 images distributed across multiple product categories. These labeled images allow the model to learn distinctive visual patterns present in product packaging, which helps

in accurately identifying products during the checkout process.

Before training, the dataset undergoes preprocessing steps such as image resizing, normalization, and dataset organization based on product classes. These preprocessing steps ensure that the images meet the input requirements of the machine learning model and improve training efficiency. The prepared dataset is then used to train and evaluate the product recognition model used in the Smart Retail Commerce System.

3.4 Proposed Methodology

The proposed methodology describes the workflow of the Smart Retail Commerce System, which integrates computer vision, barcode scanning, and machine learning techniques to automate product identification, billing, and retail analytics.

1. Image Acquisition

The process begins with capturing product images during checkout using the mobile application camera. The shopkeeper scans the product packaging and the system captures real-time frames through the device camera. These images serve as the input for the product recognition system.

2. Image Recognition using EfficientNet-B0

The captured image is processed by the image recognition module. A trained EfficientNet-B0 deep learning model analyzes the visual features present in the product packaging and predicts the corresponding product class. EfficientNet-B0 is selected due to its balance between accuracy and computational efficiency, making it suitable for real-time mobile-based recognition.

3. Barcode Scanning (Fallback Mechanism)

If the image recognition model produces a low confidence score or fails to identify the product correctly, the system activates barcode scanning as a fallback method. The barcode present on the product packaging is detected and decoded, and the product information is retrieved from the database to ensure reliable identification.

4. Transaction Recording

Once the product is successfully identified, the system records the transaction and stores it in the database. Each transaction includes product details, quantity, and timestamp, allowing the system to maintain a digital record of all sales.

5. Inventory Update

After the transaction is recorded, the inventory management module automatically updates the stock levels by deducting the sold quantity. This helps retailers maintain accurate inventory records and monitor product availability.

6. Demand Forecasting

The historical sales data stored in the database is analyzed using the ARIMA time-series forecasting model. This model helps predict future product demand based on past sales patterns, allowing retailers to manage inventory more efficiently.

7. Analytics Dashboard

The insights generated from sales data are displayed through analytics dashboards. Retailers can monitor sales trends, product performance, and inventory levels, while manufacturers can access aggregated analytics to understand regional demand patterns and market performance.

3.5 User Interface Design

The User Interface (UI) of the Smart Retail Commerce System is designed to provide a simple, intuitive, and efficient interaction for retail shopkeepers and manufacturers. The interface is developed primarily for mobile devices so that shopkeepers can easily scan products, manage transactions, and view sales insights directly from the application. The design focuses on minimizing complexity and ensuring that users with limited technical knowledge can operate the system effectively.

The main interface of the application includes a product scanning screen where the shopkeeper can capture product images using the device camera. The system processes the captured image in real time and displays the identified product along with relevant details such as product name and price. In cases where image recognition fails, the interface allows the user to activate barcode scanning as an alternative method for product identification. Once the product is confirmed, the item is added to the billing list displayed on the screen.

The application also provides a billing interface where scanned products are listed along with their quantities and prices. The system automatically calculates the total bill and records the transaction once the purchase is confirmed. This interface allows shopkeepers to complete the checkout process quickly while maintaining accurate digital records of sales transactions.

In addition to billing functionality, the system includes an analytics dashboard for retailers. This dashboard displays visual insights such as sales trends, product performance, and inventory status. These visualizations help shopkeepers understand which products are selling quickly and assist them in making informed inventory decisions.

A separate manufacturer dashboard is also provided to present aggregated sales insights collected from multiple retail stores. Through this interface, manufacturers can analyze regional demand patterns, monitor product performance, and gain a better understanding of market trends. The overall user interface design ensures that both retailers and manufacturers can easily access the information they need while interacting with the system efficiently.

3.6 Database Design

The Smart Retail Commerce System uses a relational database implemented in MySQL to store and manage information related to retail stores, transactions, product inventory, sales records, and manufacturer data. The database is designed to ensure data consistency, efficient transaction handling, and scalability as the number of stores and transactions increases. Each table in the database is structured with primary keys, foreign keys, and indexing mechanisms to maintain relationships between different entities in the system.

The shops table stores information about registered retail stores using the system. It includes details such as store name, phone number, password hash for authentication, and location information including country, state, and district. Each shop is assigned a unique identifier which is used to associate transactions, inventory records, and sales data with the corresponding store. The schema for the shops table is shown in Table 3.1.

The transactions table records billing sessions generated during the checkout process. Each transaction is linked to a specific shop through a foreign key relationship. The table stores the transaction code, total amount, transaction status, and timestamp information.

This structure allows the system to track ongoing and completed billing transactions. The transactions table schema is presented in Table 3.2.

The sales table maintains detailed records of products sold during each transaction. It stores information such as product name, category, price, quantity, total value, and transaction reference. The table is linked to the transactions table through a foreign key relationship, enabling the system to group multiple product sales under a single billing transaction. The sales table schema is described in Table 3.3.

The inventory table manages the stock information for each shop. It contains product details including product name, category, price, available stock, and barcode information. The inventory is linked to the shop through a foreign key relationship, ensuring that each shop maintains its own independent product inventory. A composite unique key ensures that the same barcode cannot be duplicated within the same shop. The inventory table schema is given in Table 3.4.

The database also includes manufacturer-related tables to support analytics and market insights. The manufacturers table stores details about registered manufacturing companies including company name, contact information, and authentication credentials, as shown in Table 3.5. The manufacturer_products table maintains a list of products produced by each manufacturer and links them through a foreign key relationship, as shown in Table 3.6. This structure enables the system to associate product performance data with the corresponding manufacturer.

Overall, the database design supports efficient management of retail operations including product identification, billing transactions, inventory updates, and sales analytics. The relational structure ensures data integrity while enabling efficient queries for generating insights used in retailer dashboards and manufacturer analytics.

Table 3.1: Shops Table Schema

Field Name	Data Type	Description
id	INT (PK)	Unique identifier for each shop
store_name	VARCHAR(255)	Name of the retail store
phone	VARCHAR(20)	Unique phone number used for login
password_hash	VARCHAR(255)	Encrypted password for authentication
country	VARCHAR(100)	Country where the shop is located
state	VARCHAR(100)	State where the shop operates
district	VARCHAR(100)	District location of the shop
created_at	TIMESTAMP	Timestamp of shop registration

Table 3.2: Transactions Table Schema

Field Name	Data Type	Description
id	INT (PK)	Unique transaction identifier
shop_id	INT (FK)	References the shop performing the transaction
transaction_code	VARCHAR(20)	Unique code generated for each billing session
total	DECIMAL(10,2)	Total amount of the transaction
status	ENUM	Transaction status (active or completed)
created_at	DATETIME	Timestamp when the transaction was created

Table 3.3: Sales Table Schema

Field Name	Data Type	Description
id	INT (PK)	Unique identifier for each sale record
shop_id	INT (FK)	Identifier of the shop where the sale occurred
product_name	VARCHAR(255)	Name of the sold product
category	VARCHAR(100)	Product category
price	DECIMAL(10,2)	Price of a single unit of the product
quantity	INT	Number of units sold
total	DECIMAL(10,2)	Total value of the sale
transaction_id	INT (FK)	Reference to the associated transaction
created_at	TIMESTAMP	Timestamp when the sale was recorded

Table 3.4: Inventory Table Schema

Field Name	Data Type	Description
id	INT (PK)	Unique identifier for inventory records
shop_id	INT (FK)	Identifier of the shop owning the inventory
product_name	VARCHAR(255)	Name of the product
category	VARCHAR(100)	Category of the product
price	DECIMAL(10,2)	Selling price of the product
stock	INT	Available stock quantity
barcode	VARCHAR(50)	Barcode associated with the product
created_at	TIMESTAMP	Timestamp when the product was added

Table 3.5: Manufacturers Table Schema

Field Name	Data Type	Description
id	INT (PK)	Unique identifier for each manufacturer
company_name	VARCHAR(255)	Name of the manufacturing company
email	VARCHAR(255)	Unique email address of the manufacturer
phone	VARCHAR(20)	Contact phone number
password_hash	VARCHAR(255)	Encrypted password for authentication
country	VARCHAR(100)	Country where the manufacturer is located
state	VARCHAR(100)	State of the manufacturer
created_at	TIMESTAMP	Timestamp when the manufacturer account was created

Table 3.6: Manufacturer Products Table Schema

Field Name	Data Type	Description
id	INT (PK)	Unique identifier for manufacturer products
manufacturer_id	INT (FK)	References the manufacturer producing the product
product_name	VARCHAR(255)	Name of the product manufactured
category	VARCHAR(100)	Category of the product
created_at	TIMESTAMP	Timestamp when the product entry was created

3.7 Description of Implementation Strategies

The Smart Retail Commerce System is implemented using a combination of computer vision, database management, and machine learning techniques to automate retail billing and analytics. The implementation strategy focuses on integrating multiple modules that work together to provide real-time product identification, transaction management, inventory tracking, and sales analytics.

The system is developed with a mobile-based interface that allows shopkeepers to scan product packaging using the device camera. The captured images are processed by the image recognition module, which utilizes the EfficientNet-B0 deep learning model to classify the product based on visual features of its packaging. EfficientNet-B0 is selected because of its high accuracy and computational efficiency, making it suitable for real-time applications on devices with limited computational resources.

To improve reliability, a barcode scanning mechanism is implemented as a fallback strategy. When the image recognition model produces a low confidence score or fails to identify a product, the barcode scanning module is activated to decode the barcode and retrieve product information from the database. This hybrid approach ensures accurate product identification even in challenging conditions such as poor lighting or visually similar packaging.

The backend of the system is implemented using a relational database that manages information related to shops, inventory, transactions, and sales records. The database structure is designed with appropriate primary keys, foreign keys, and indexing to maintain data consistency and enable efficient retrieval of information. Each transaction recorded in the system automatically updates the corresponding inventory levels, ensuring that stock information remains accurate.

In addition to transaction management, the implementation also includes a data analytics component that processes historical sales data. A time-series forecasting model based on ARIMA is applied to analyze past sales patterns and predict future product demand. These predictions help retailers optimize inventory management and prevent stock shortages.

The system also includes interactive dashboards for both retailers and manufacturers. Retailers can monitor sales trends, product performance, and inventory levels through

a shopkeeper dashboard, while manufacturers can access aggregated analytics through a manufacturer dashboard to analyze product demand across different regions. This integrated implementation strategy ensures efficient retail operations while providing valuable insights for data-driven decision making.

3.8 Module Division

The system is divided into several functional modules to ensure organized development and efficient implementation of the Smart Retail Commerce System.

Module 1: Dataset Preparation and Image Recognition

This module focuses on collecting and preparing product image data used for training the image recognition model. The dataset is cleaned, labeled, and resized to ensure consistency during model training. A deep learning model based on EfficientNet-B0 is trained to recognize different retail product packages. The model is tested and optimized to ensure accurate and efficient product identification during real-time scanning.

Module 2: Barcode Scanning and ARIMA Integration

This module implements a barcode scanning mechanism that acts as a fallback when image recognition fails or produces low confidence results. The system decodes barcode values and retrieves corresponding product information from the database. In addition, this module integrates the ARIMA time-series model to analyze historical sales data and support demand forecasting.

Module 3: Mobile Application, Sales Logging and Bookkeeping

This module focuses on developing the mobile application interface used by shopkeepers. It integrates the device camera for product scanning and records sales transactions automatically. The system maintains digital bookkeeping records, including timestamped sales data and daily transaction summaries for efficient retail management.

Module 4: Business Insights and Forecasting

This module analyzes the collected sales data to generate business insights. It identifies patterns such as fast-moving and slow-moving products, peak sales periods, and potential stock shortages. Forecasting techniques are applied to predict future demand, and the results are displayed through analytics dashboards for retailers and manufacturers.

3.9 Work Schedule - Gantt Chart

The project was carried out over a period from December 2025 to March 2026. The schedule included several phases such as dataset collection and preparation, training of the image recognition model, development of the mobile application interface, system integration, and final testing. The overall project timeline and task distribution across modules are illustrated in the Gantt chart shown in Figure 3.4.

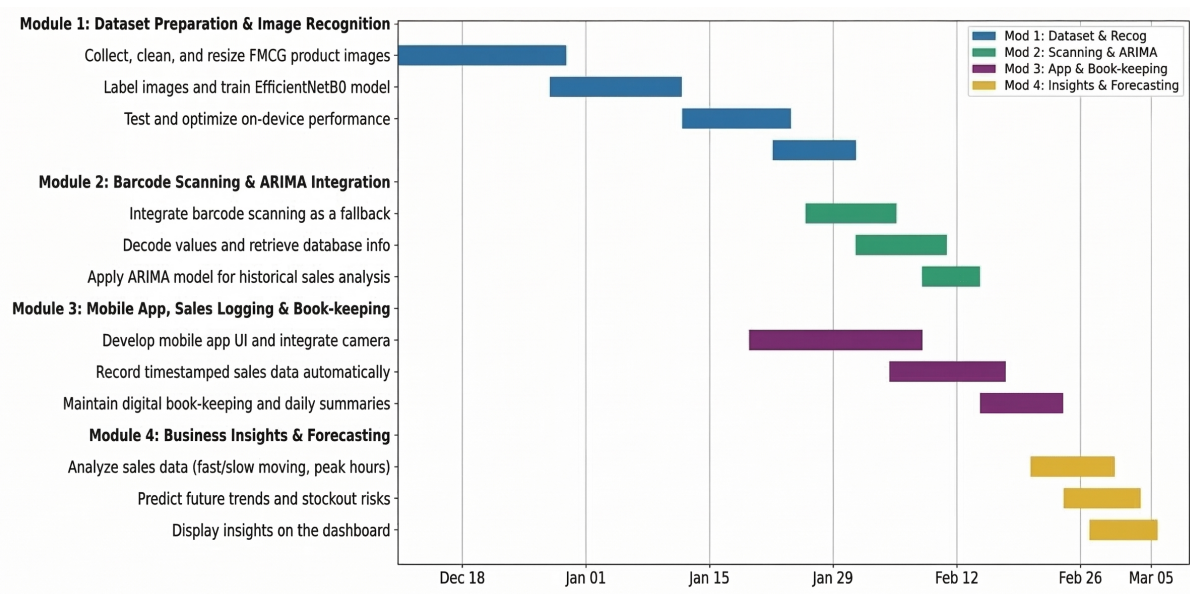


Figure 3.4: Project Work Schedule

Chapter 4

Results and Discussions

4.1 Overview

This chapter presents the results obtained from the implementation of the Smart Retail Commerce System and provides a discussion of the system's performance and effectiveness. The results evaluate the functionality of different modules including product recognition, barcode scanning, transaction management, inventory tracking, and sales analytics.

The implemented system successfully integrates computer vision and machine learning techniques to automate product identification and retail billing. The EfficientNet-B0 model was used for image-based product recognition, enabling the system to classify retail product packages with high accuracy. In cases where image recognition produced low confidence results, the barcode scanning module acted as a reliable fallback mechanism to ensure correct product identification.

The system was also evaluated based on its ability to record transactions, update inventory automatically, and store sales data for further analysis. The collected sales data was processed using the ARIMA time-series model to generate demand forecasts and identify sales trends. The results demonstrate that the system can effectively support retail operations by reducing manual effort and providing useful business insights through analytics dashboards.

Overall, the experimental results show that the proposed system is capable of performing real-time product recognition, maintaining accurate sales records, and generating meaningful analytical insights. These results validate the feasibility of using computer vision and machine learning techniques to enhance retail management systems.

4.2 Testing

System testing was carried out to verify the functionality and performance of the Smart Retail Commerce System. The testing process focused on validating the main features of the application, including product scanning, product identification using the EfficientNet-B0 model, barcode scanning as a fallback mechanism, transaction recording, and inventory updates. Each component of the system was tested individually and then evaluated as part of the integrated system to ensure smooth interaction between modules.

The mobile application interface was tested by performing multiple product scanning operations using the device camera. The image recognition module was evaluated based on its ability to correctly identify product packaging from the trained dataset. In situations where the model produced low confidence predictions, the barcode scanning functionality was tested to confirm that the system could still retrieve the correct product information from the database. This ensured that the system maintained reliable product identification during checkout.

Additional testing was conducted on the transaction management and inventory modules. The system successfully recorded sales transactions and updated the inventory levels automatically after each sale. The generated sales data was stored in the database and later used for analytics and demand forecasting using the ARIMA model.

This section also includes screenshots of the mobile application interface demonstrating key functionalities such as product scanning, billing interface, transaction logging, and inventory updates. In addition, screenshots of the manufacturer dashboard are presented to illustrate how aggregated sales data and product performance insights are visualized for manufacturers. These interfaces demonstrate how the system supports both retailers and manufacturers by providing real-time operational tools and analytical insights. The mobile application screens include the Sign Up page, App Dashboard, Transaction History, Image Scan Interface, Barcode Scan Interface, Manual Entry Interface, Product Confirmation page, App Analytics page, Inventory page, and Profile page. The web-based manufacturer interfaces, including the Web Dashboard, Add Product page, and Web Analytics page, are shown in Figures 4.11, 4.12, and 4.13 respectively.

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Dashboard

PK STORES

KERALA, INDIA

TODAY'S SALE

DATE: 09/03/2026 (Historical)

DESCRIPTION	QTY	RATE	AMOUNT
Exo Touch - Shine Anti...	8.00	29.00	232.00
Appy Fizz Apple Juice ...	12.00	20.00	240.00
Appy Fizz Apple Juice ...	7.00	20.00	140.00
Bisleri Packaged Drink...	15.00	20.00	300.00
Coca-Cola-1 liter-	16.00	50.00	800.00
Exo Touch - Shine Anti...	1.00	29.00	29.00

View Full Sales

Transactions

TODAY'S ANALYTICS

₹1783.00

Revenue

56

Items Sold

9

Transactions

Dashboard

Inventory

Analytics

Profile

Figure 4.1: Sign Up Page

Figure 4.2: App Dashboard

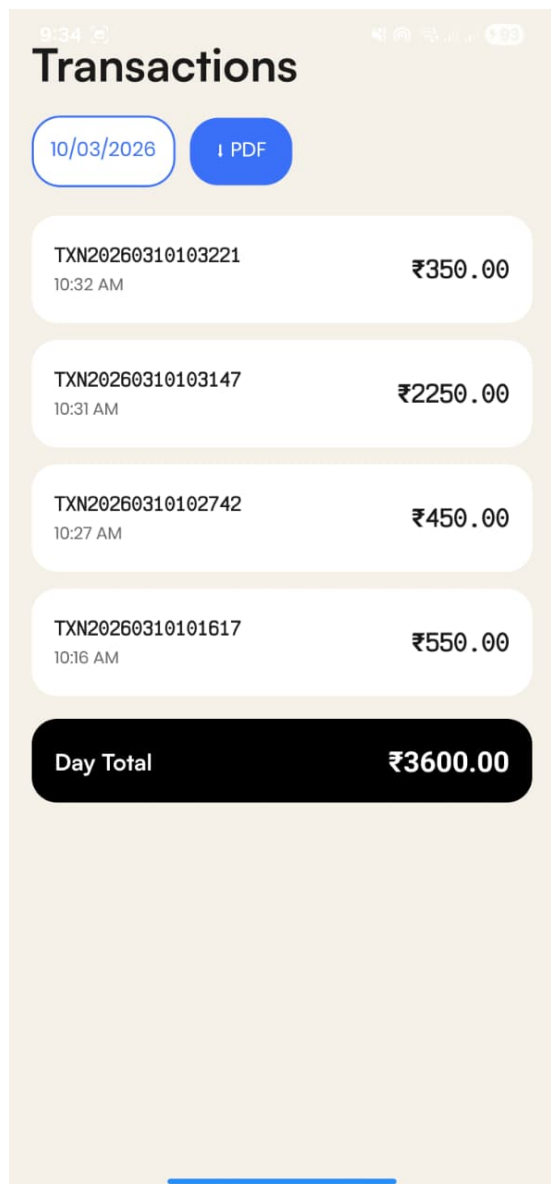


Figure 4.3: Transaction History Page



Figure 4.4: Image Scan Interface

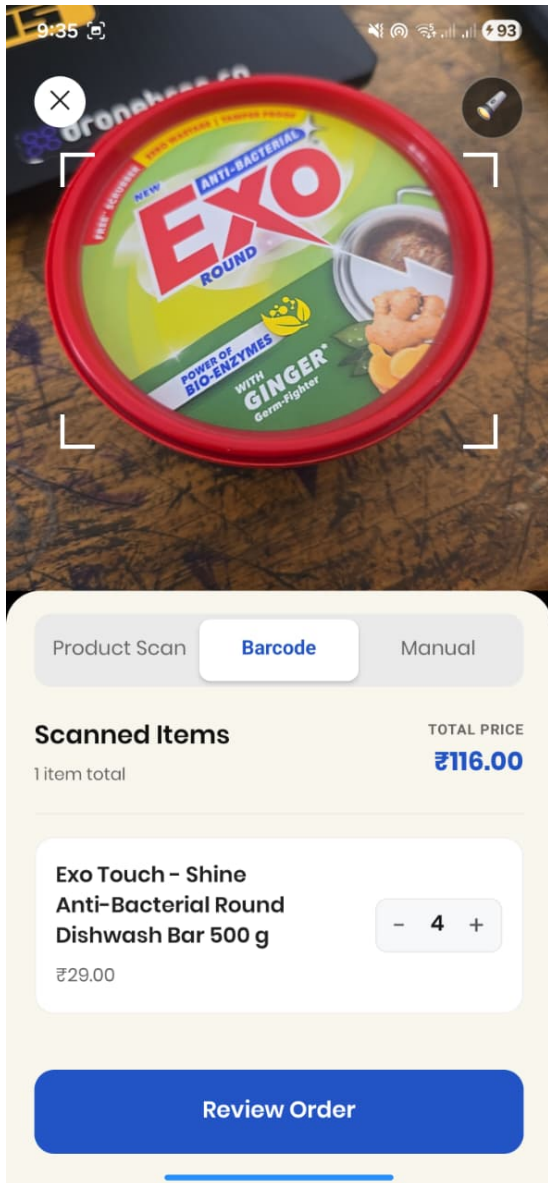


Figure 4.5: Barcode Scan Interface

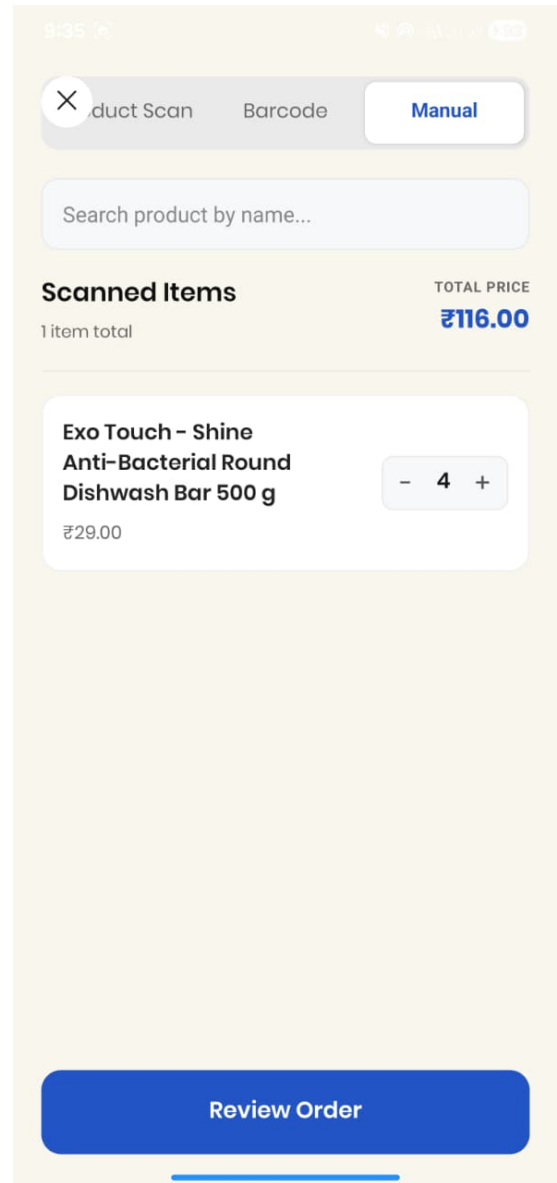


Figure 4.6: Manual Entry Interface

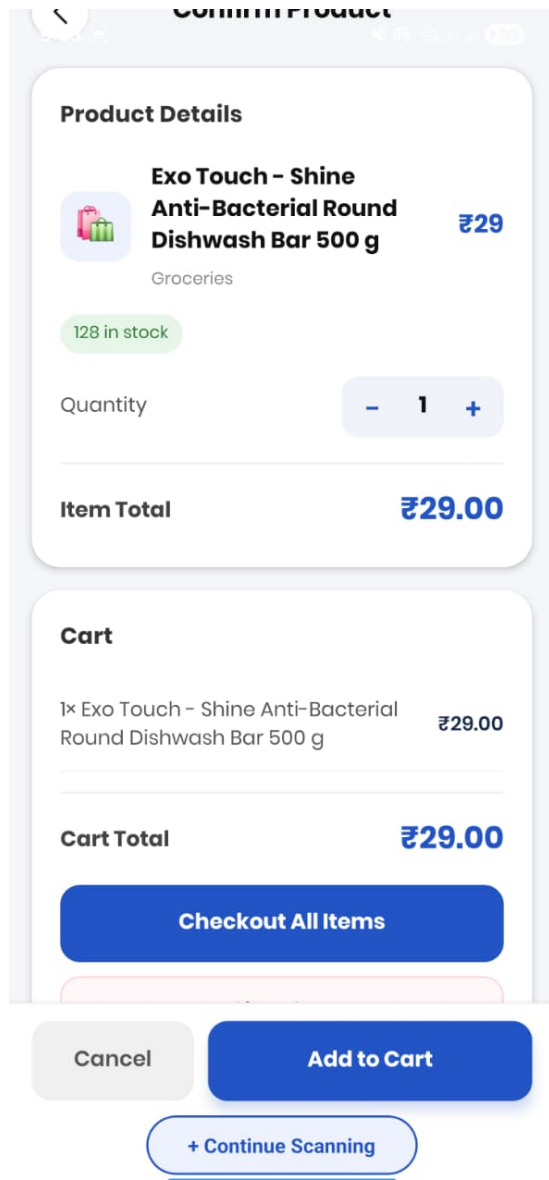


Figure 4.7: Product Confirmation Page

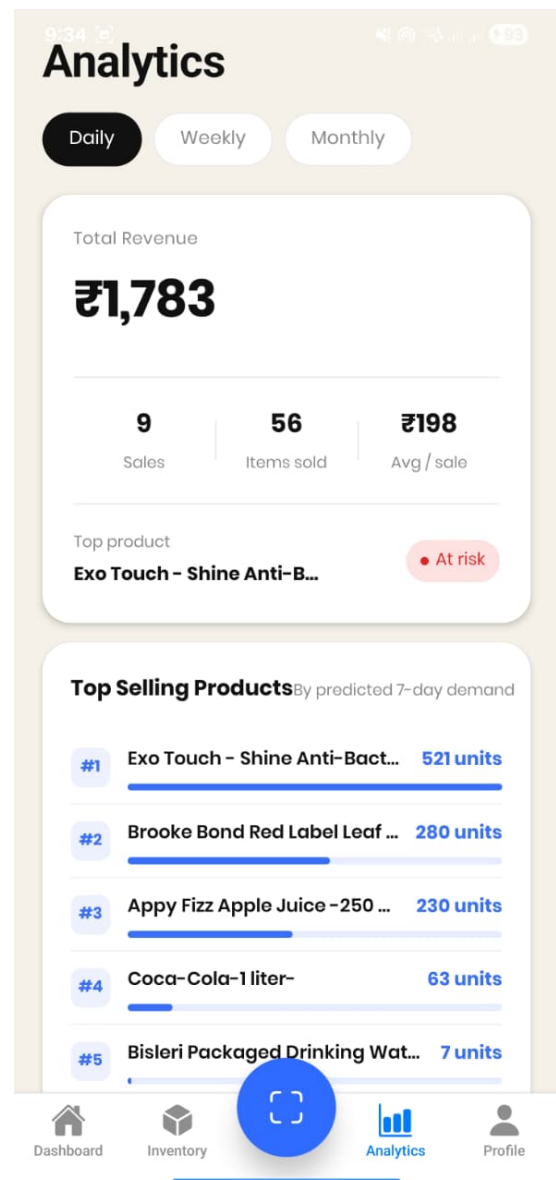


Figure 4.8: App Analytics Page

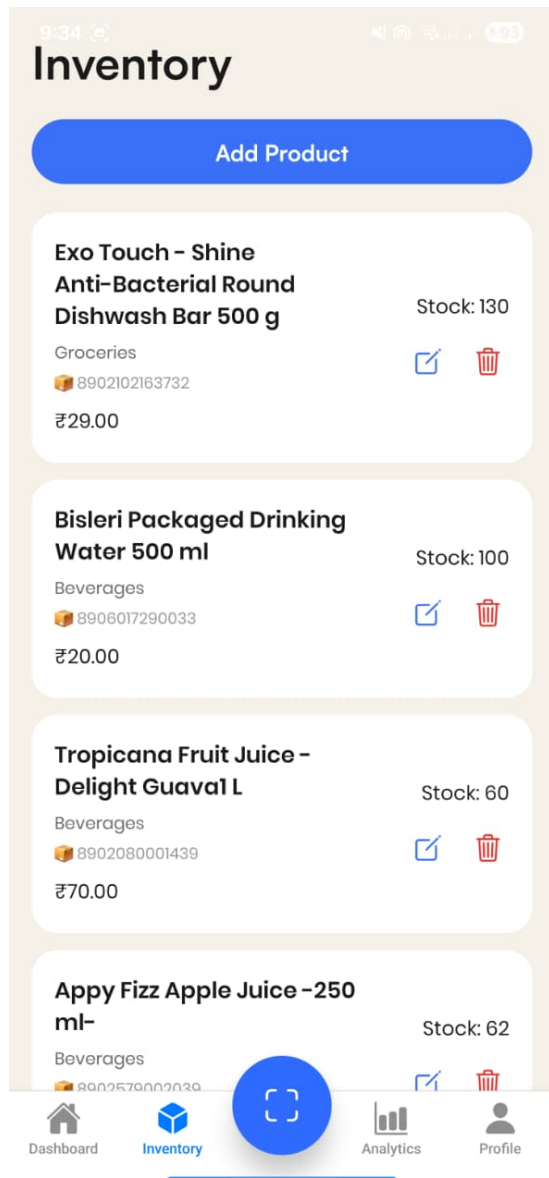


Figure 4.9: Inventory Page

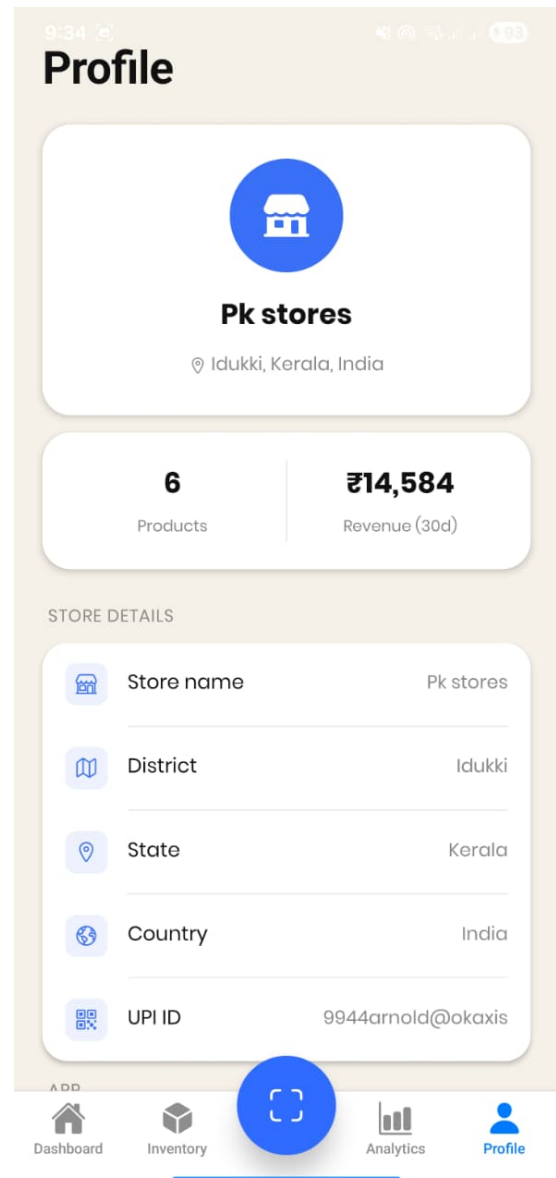


Figure 4.10: Profile Page

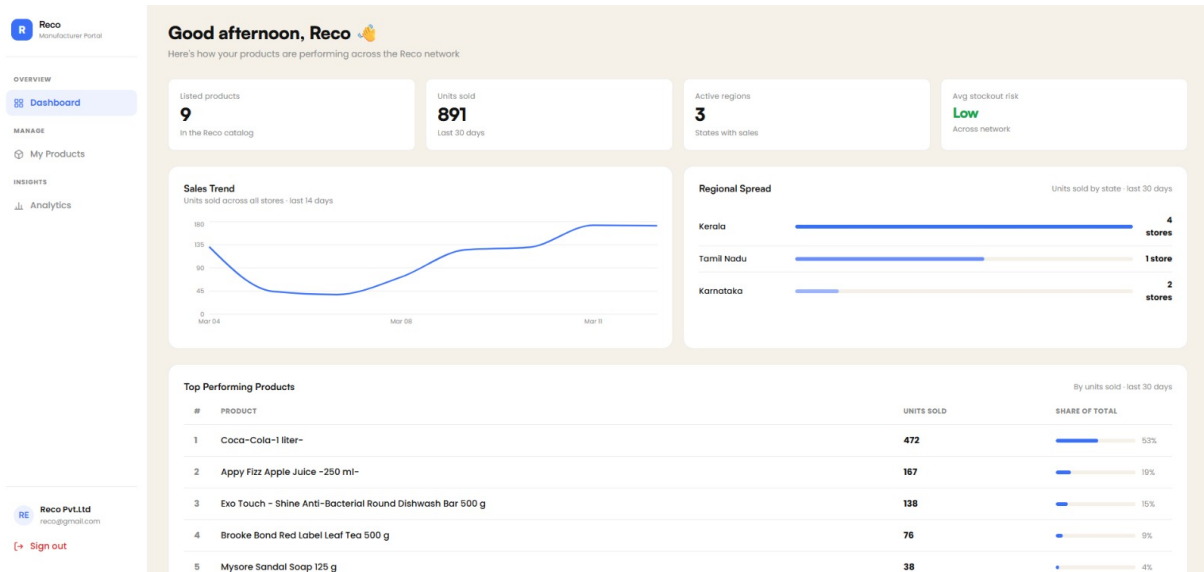


Figure 4.11: Web Dashboard

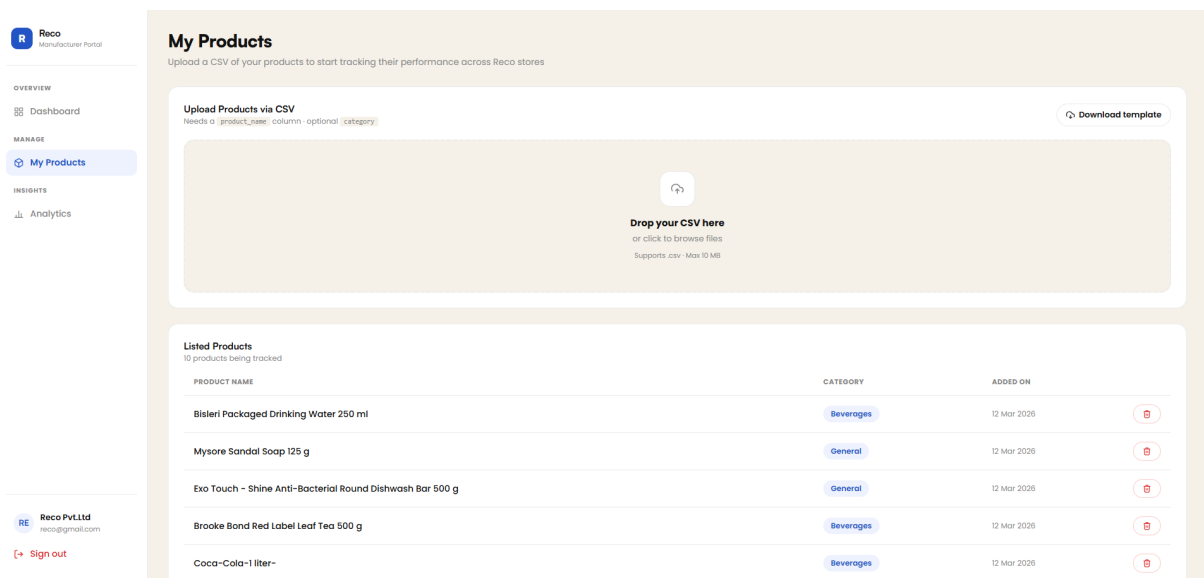


Figure 4.12: Add Product Page

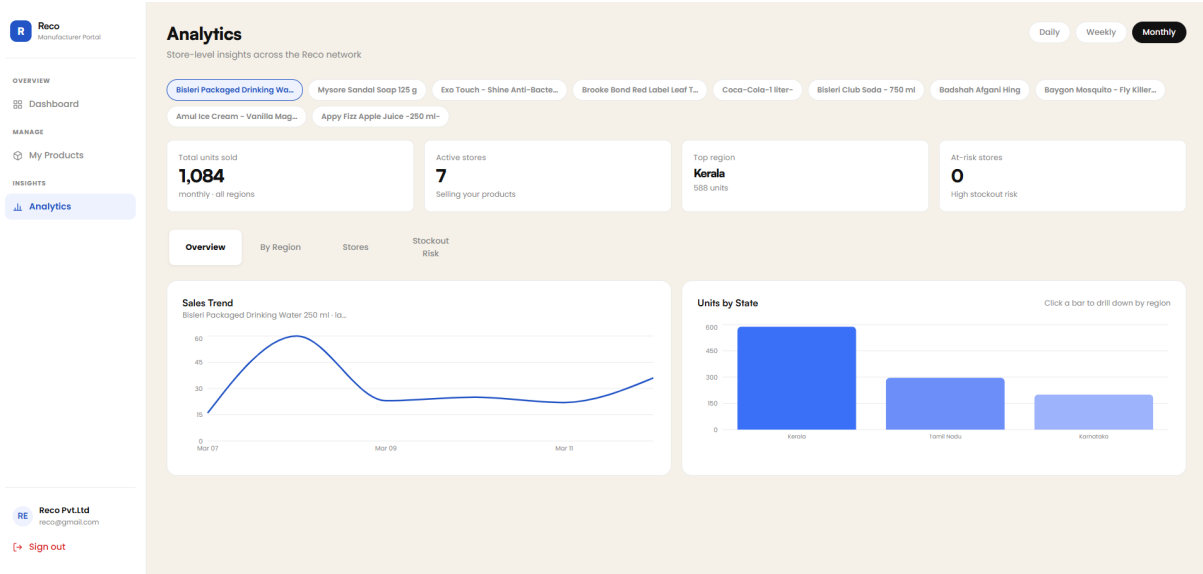


Figure 4.13: Web Analytics Page

4.3 Quantitative Results

The performance of the image recognition model based on EfficientNet-B0 was evaluated using training and validation metrics such as accuracy and top-3 accuracy. These metrics provide insights into the learning behavior of the model during training and fine-tuning. The training and validation accuracy curves are shown in Figure 4.14, and the top-3 accuracy is illustrated in Figure 4.15.

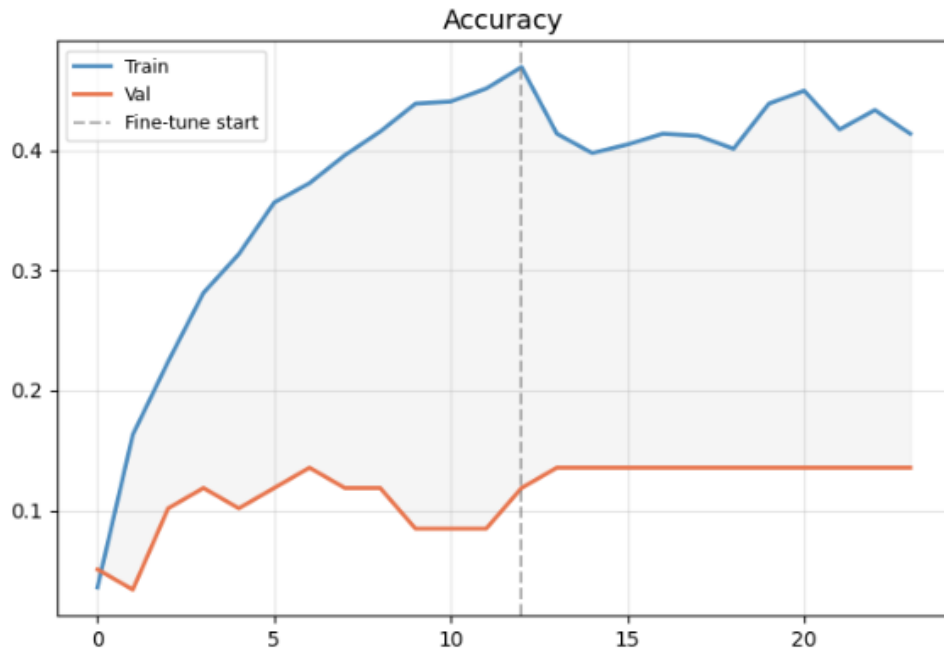


Figure 4.14: Training and Validation Accuracy

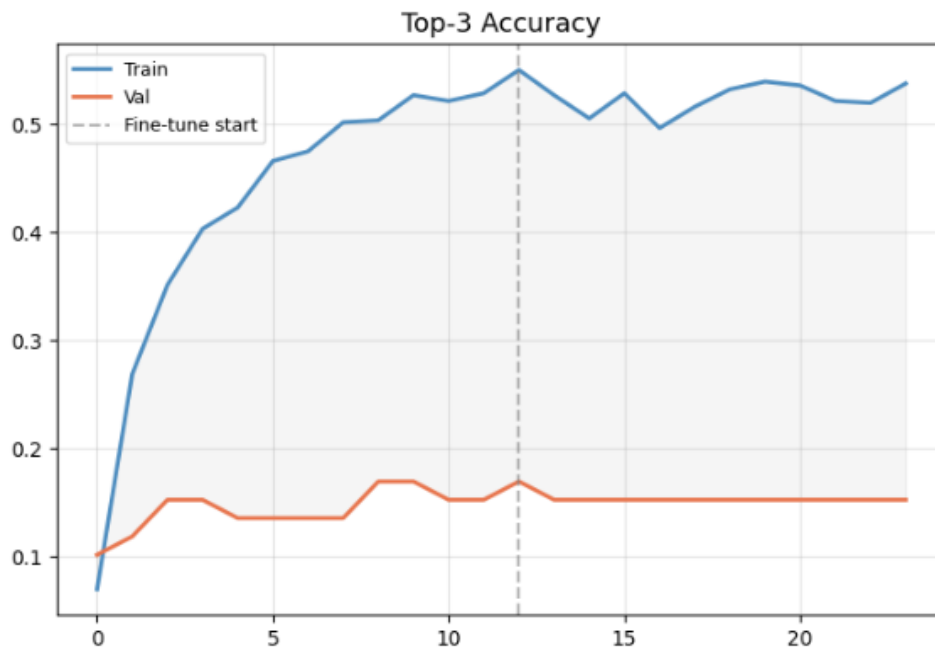


Figure 4.15: Top-3 Accuracy

The top-3 accuracy graph (Figure 4.15) shows higher values compared to the standard accuracy, indicating that the correct class is often among the top three predictions made by the model. This demonstrates that the model is able to capture relevant features of the products, even if the top prediction is not always correct. Such behavior is useful

in practical scenarios where multiple candidate predictions can assist in improving final identification.

Overall, the evaluation results indicate that the EfficientNet-B0 model is capable of learning meaningful visual features for product recognition. However, the difference between training and validation performance highlights the need for further improvements such as increasing dataset size, applying data augmentation, and tuning model parameters to enhance generalization.

4.4 Discussion

The results obtained from the implementation of the Smart Retail Commerce System demonstrate the effectiveness of integrating computer vision and machine learning techniques into retail operations. The system successfully performs real-time product identification using the EfficientNet-B0 model, which provides a good balance between accuracy and computational efficiency. The model was able to correctly classify most products from the dataset, showing its suitability for practical retail environments.

However, certain challenges were observed during testing, particularly in scenarios involving similar product packaging, poor lighting conditions, or partially visible labels. In such cases, the confidence of the image recognition model decreased. The inclusion of the barcode scanning module as a fallback mechanism proved to be effective in handling these situations, ensuring reliable product identification and minimizing errors during the billing process.

The transaction management and inventory modules functioned as expected, accurately recording sales and updating stock levels in real time. This demonstrates the system's capability to maintain consistent and reliable digital records, which is essential for retail bookkeeping. The integration of these modules reduces manual effort and minimizes human errors typically associated with traditional billing systems.

The demand forecasting component using the ARIMA model provided useful insights into sales trends and future product demand. While the predictions were generally consistent with historical data patterns, the accuracy of forecasting depends on the availability of sufficient and high-quality data. As more transaction data is collected over time, the forecasting performance can be further improved.

The user interfaces for both the mobile application and the manufacturer dashboard were found to be intuitive and effective in presenting system functionalities and analytical insights. Retailers can easily perform product scanning and billing operations, while manufacturers can analyze aggregated sales data to understand market trends and product performance.

Overall, the system demonstrates strong potential for improving retail efficiency by automating billing processes and providing data-driven insights. Future improvements could include enhancing the robustness of the image recognition model, expanding the dataset to include more product variations, and incorporating more advanced forecasting techniques to improve prediction accuracy.

Chapter 5

Conclusion

5.1 Conclusion

The Smart Retail Commerce System demonstrates the effective integration of computer vision and machine learning to enhance retail operations. By utilizing the EfficientNet-B0 model for image-based product recognition along with barcode scanning as a fallback mechanism, the system automates billing, reduces manual effort, and improves accuracy. This results in a faster and more efficient checkout process compared to traditional retail systems.

In addition, the system incorporates inventory management and data-driven analytics to provide valuable insights such as product performance and demand trends. The ARIMA-based forecasting module further supports decision-making by predicting future demand. Overall, the system is reliable, scalable, and suitable for real-world applications, offering a cost-effective solution for modern retail management.

5.2 Future Scope

The system can be further enhanced by improving the accuracy and robustness of the image recognition model through dataset expansion and advanced model tuning techniques. Implementing multi-product detection using real-time object detection models such as YOLO can significantly improve efficiency by allowing multiple items to be scanned simultaneously.

Future improvements can also include the use of advanced forecasting models such as LSTM for better prediction accuracy, along with cloud integration for real-time data synchronization across multiple stores. Additionally, features such as voice assistance, multilingual support, and automated restocking alerts can improve usability and expand the system's practical applicability.

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Appendix A: Presentation

Retail Commerce using Image Recognition and ML

Final Presentation

Abhishikth S Mattom
U2303011

Alan Jophy
U2303021

Alen Abraham Saji
U2303027

Arnold Shibu
U2303052

Guide
Dr. Amitha Mathew

1

Introduction

- Retail commerce in India generates large amounts of sales data but lacks effective analytical tools.
- Small retailers depend on manual or basic digital methods that are slow and error-prone.
- The application uses image recognition to automate product identification during checkout.
- Machine learning converts sales data into insights for inventory and demand management.
- Anonymized analytics also help manufacturers understand region-wise market demand.

2



Problem Definition

The system automates product identification during checkout and provides data-driven insights for shopkeepers and manufacturers.

3

Objectives

- | | |
|--------------------------------------|--|
| Automated Product Recognition | - Automatically identify products and records sales at the time of purchase. |
| Sales Tracking | - Automated bookkeeping and sales tracking |
| Sales Trends and Product Performance | - Provide live updates of product-wise sales performance. |
| Inventory and Demand Forecasting | - Predict demand and alerts about potential stockouts. |
| Manufacturer Market Intelligence | - Reveal overall market demand and regional sales patterns. |

4

Scope and Relevance

Scope

- Smart retail checkout using image recognition (EfficientNetB0)
- Automatic product identification and sales recording
- Barcode scanning fallback if recognition fails

Relevance

- Reduces manual billing errors
- Improves checkout speed
- Enables retailers make data-driven inventory decisions

Applications

- Retail shopkeepers – Faster billing and better inventory control with improved sales tracking and forecasting
- Manufacturers – Insights into regional demand and product performance

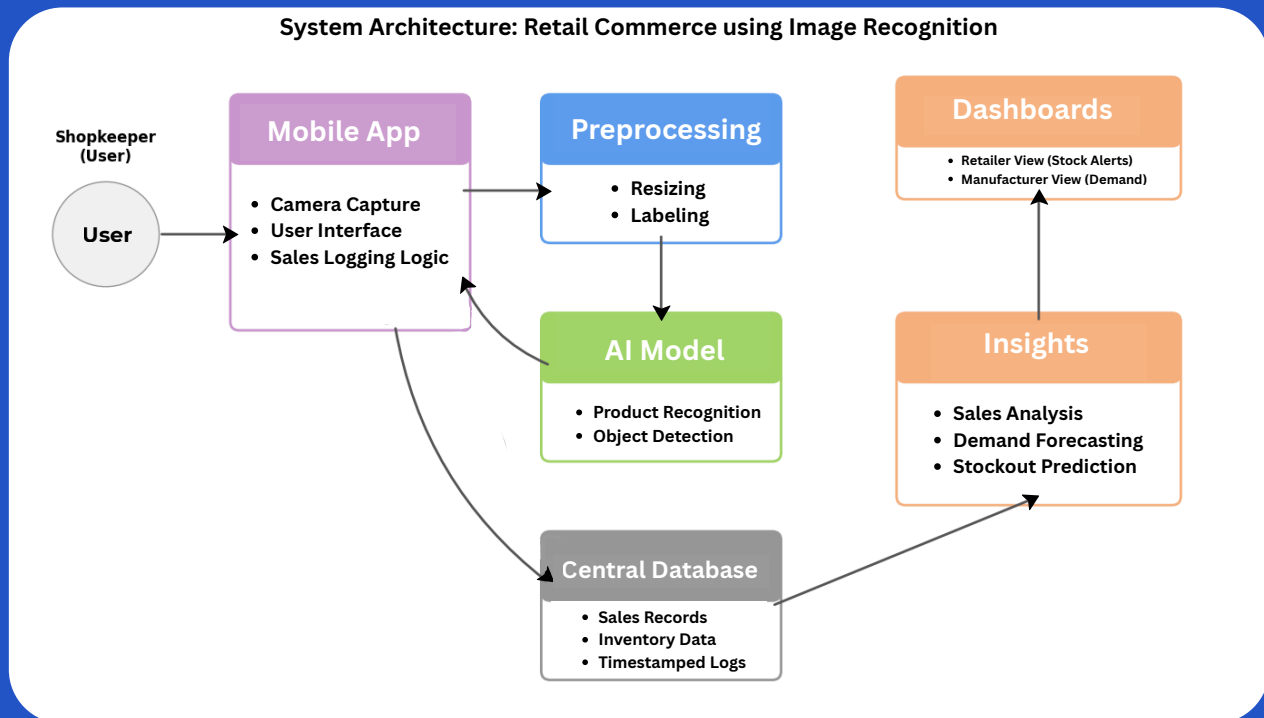
5

System Design

- Shopkeeper scans product packaging during checkout using the mobile application.
- Image recognition model processes the image on-device and identifies the product instantly.
- Identified product and sale details are stored as timestamped sales data.
- Machine learning engine analyzes sales data to generate insights and demand forecasts.
- Retail insights are shown in the mobile app, while aggregated analytics are delivered to manufacturers through a web dashboard.

6

Architectural Design



7

Module 1: Dataset Preparation & Image Recognition

- Collect Indian FMCG product images from public datasets and online sources
- Clean and resize images to standard input dimensions
- Label images with product names
- Train EfficientNetB0 model for product image recognition
- Test and evaluate model accuracy
- Optimize model for on-device performance

Handled By : Alan Jophy, Abhishikth S Mattom

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Module 2: Barcode scanning and ARIMA integration

- Integrate barcode scanning when image recognition fails.
- Decode barcode values and retrieve product information from database.
- Apply ARIMA model for sales demand forecasting.
- Analyse historical sales data to predict future product demand.

Handled By : Alen Abraham Saji , Arnold Shibu

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Module 3: Mobile App, Sales Logging & Book-keeping

- Develop mobile app for product scanning
- Integrate camera and backend services
- Record timestamped sales data automatically
- Maintain digital book-keeping records for daily transactions
- Generate daily sales summaries

Handled By : Abhishikth S Mattom , Arnold Shibu

10

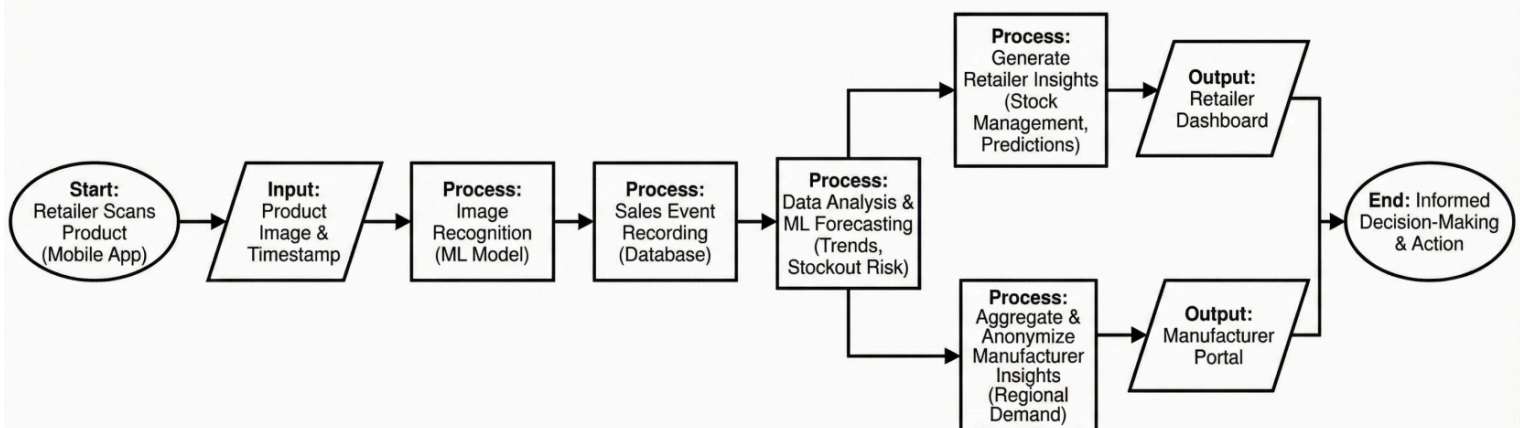
Module 4: Business Insights & Forecasting

- Analyze sales data using ML techniques
- Identify fast/slow moving products
- Detect and display peak sales hours based on transaction timestamps
- Predict future sales trends
- Predict stock requirements and stockout risks
- Display insights on dashboard

Handled By : Alan Jophy , Alen Abraham Saji , Arnold Shibu , Abhishikth S Mattom

11

Flow Diagram: Smart Product Recognition & Business Insights System



12

Dataset

Dataset Name: Retail Store Indian Trial – Grocery Item Images

Source: <https://universe.roboflow.com/nmims-b6ta9/retail-store-indian-trial>

Description:

- A labeled dataset of Indian retail product images.
- Contains multiple product categories relevant to grocery and FMCG goods.

Purpose: Used to train and fine-tune the image recognition model for identifying products from their packaging during checkout.

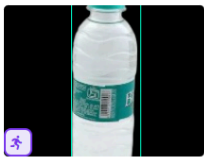
Key Features:

- Multiple images per product with variations in angle and lighting.
- Contains Indian product packaging examples.

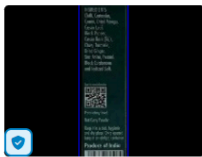
Image Count: 754 images

13

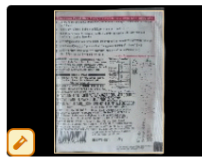
Sample Images



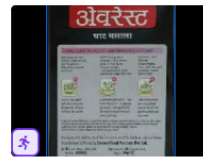
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everest-pav-bhaji-masala-1...



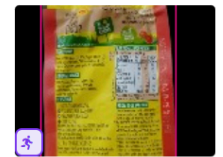
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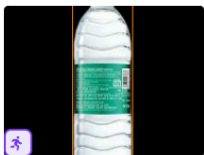
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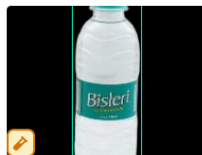
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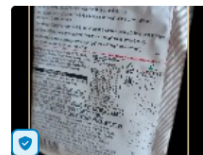
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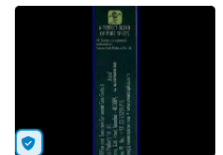
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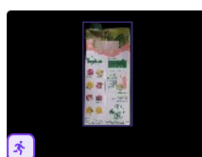
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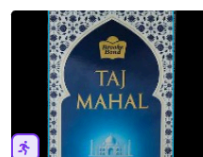
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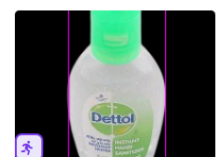
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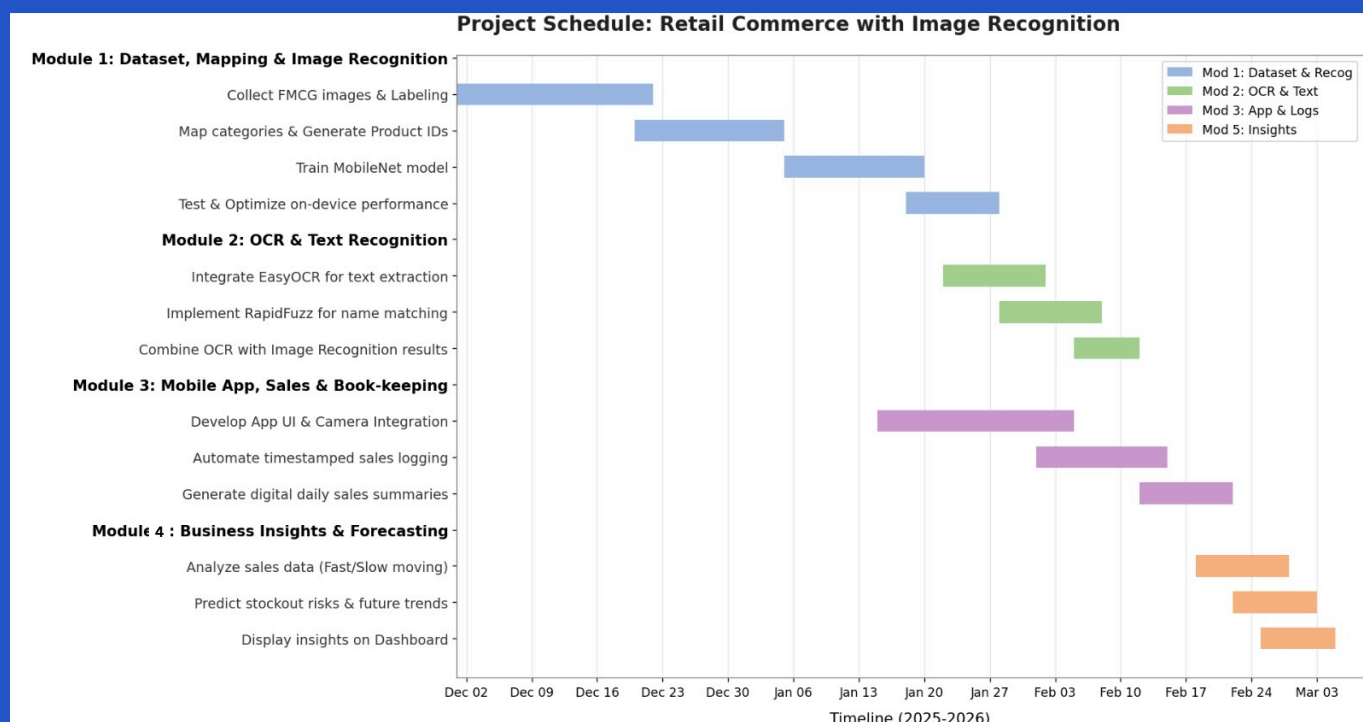
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14

Work Division



15

Software/ Hardware Requirements

Software

Programming Languages: Python, JavaScript

Machine Learning Models:

- **EfficientNetB0** – for on-device product image recognition
- **ARIMA** – for sales demand forecasting

Database: MySQL

Cloud Platform: AWS (Backend, Analytics, Storage)

Hardware

Mobile Device: Android smartphone with camera (checkout scanning)

Processor: Intel i5 / AMD Ryzen 5 or higher

RAM: Minimum 8 GB

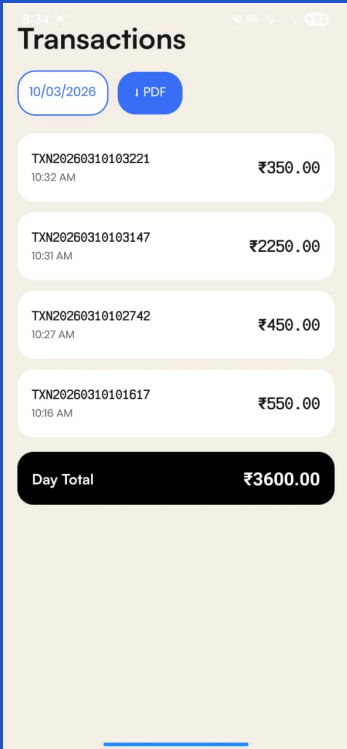
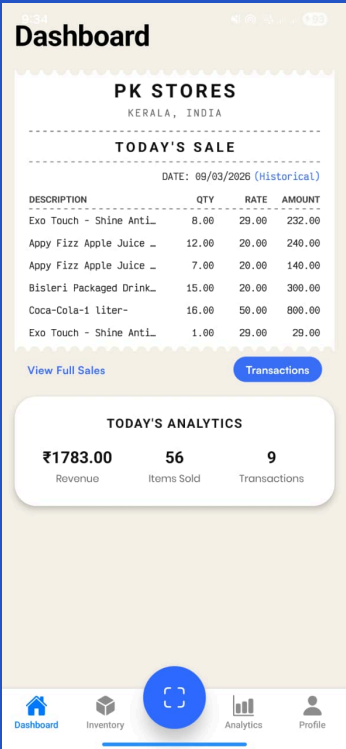
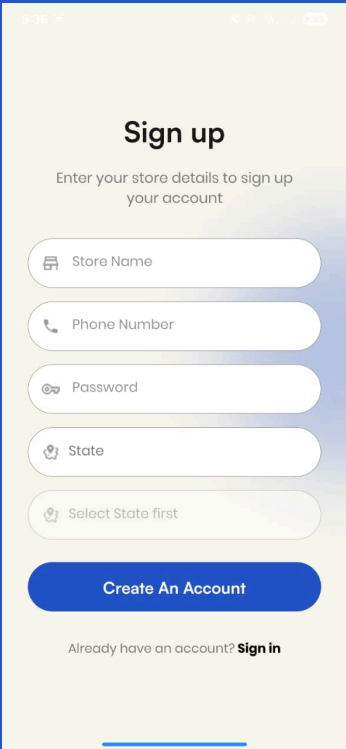
Storage: Minimum 256 GB

Internet: Required for data synchronization and analytics

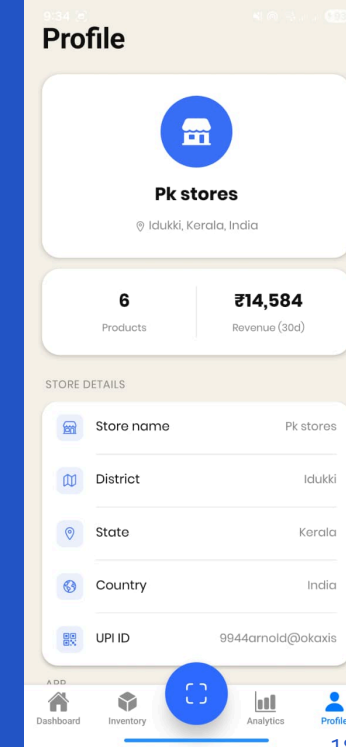
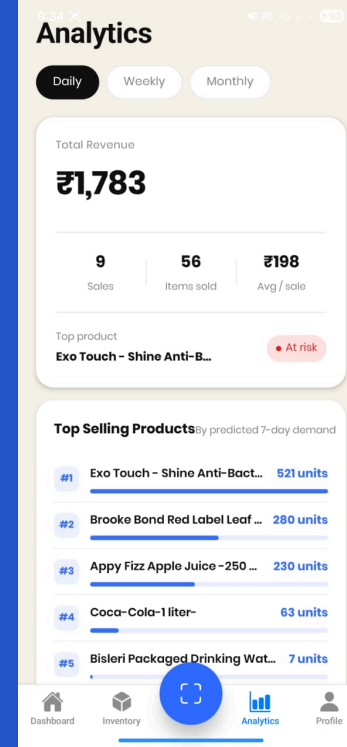
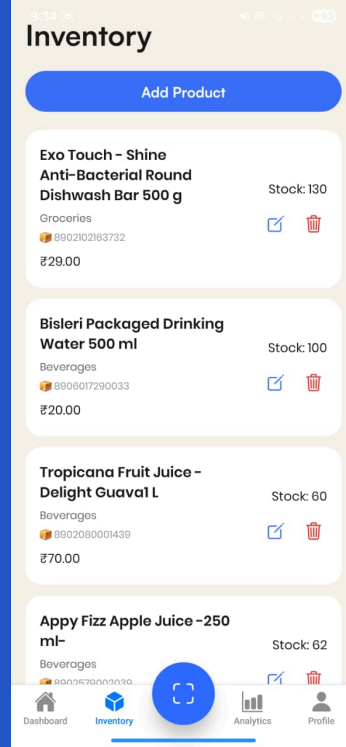
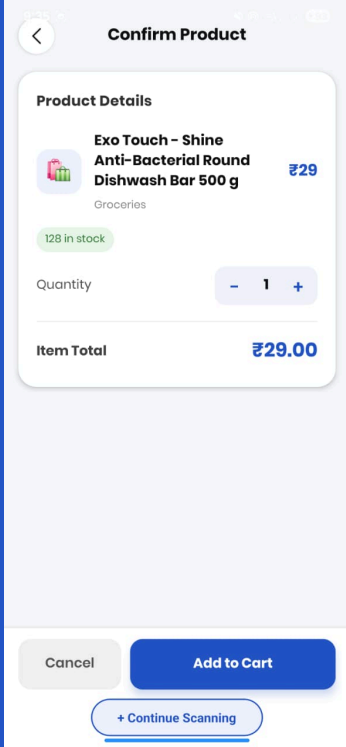
Optional: GPU (for faster model training)

16

Results - Mobile App



Results - Mobile App



Future Enhancements

- Multi-product detection to scan multiple items in a single frame
- Supplier integration for automatic product restocking
- Mobile-based POS replacement with billing machine integration to reduce hardware cost

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Thank You

Appendix B:

Vision, Mission, POs, PSOs and COs

Vision, Mission, POs, PSOs and COs

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

4. Conduct Investigations of Complex Problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the responsibilities relevant to professional engineering practice.

7. Environment and Sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

9. Individual and Team Work: Function effectively as an individual and as a member or leader in teams and multidisciplinary settings.

10. Communication: Communicate effectively with the engineering community and society at large. Be able to comprehend and write effective reports and documentation, make effective presentations, and give and receive clear instructions.

11. Project Management and Finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, and manage projects in multidisciplinary environments.

12. Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Programme Specific Outcomes (PSO)

PSO1: Computer Science Specific Skills

The ability to identify, analyze, and design solutions for complex engineering problems in multidisciplinary areas by understanding core computer science principles.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software development.

PSO3: Professional Skills

The ability to apply computer science fundamentals in research and develop innovative products to meet societal needs.

Appendix C:

CO-PO-PSO Mapping

COURSE OUTCOMES:

Sl.No	DESCRIPTION	Blooms' Taxonomy Level
CO1	Identify technically and economically feasible problems.	Apply
CO2	Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes.	Apply
CO3	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced programming techniques.	Apply
CO4	Prepare technical report and deliver presentation.	Apply
CO5	Apply engineering and management principles to achieve the goal of the project.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO5	PO6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	2	2	3	2	2	2	3	2	2	2
CO2	3	3	3	3	3	-	3	2	3	2	3	2	2	2
CO3	3	3	3	3	3	2	3	2	2	2	3	-	-	2
CO4	2	3	2	2	2	-	3	3	3	2	3	2	2	2
CO5	3	3	3	2	2	2	3	2	-	2	3	2	2	2

JUSTIFICATIONS FOR CO-PO MAPPING

MAPPING	LOW/MEDIUM /HIGH	JUSTIFICATION
101003/CS6 22T.CO1- PO1	HIGH	Identify technically and economically feasible problems by applying the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
101003/CS6 22T.CO1- PO2	HIGH	Identify technically and economically feasible problems by analysing complex engineering problems reaching substantiated conclusions using first principles of mathematics.
101003/CS6 22T.CO1- PO3	HIGH	Design solutions for complex engineering problems by identifying technically and economically feasible problems
101003/CS6 22T.CO1- PO4	HIGH	Identify technically and economically feasible problems by analysis and interpretation of data
101003/CS6 22T.CO1- PO5	MEDIUM	Responsibilities relevant to the professional engineering practice by identifying the problem.
101003/CS6 22T.CO1- PO6	MEDIUM	Identify technically and economically feasible problems by understanding the impact of the professional engineering solutions
101003/CS6 22T.CO1- PO7	HIGH	Apply ethical principles and commit to professional ethics to identify technically and economically feasible problems
101003/CS6 22T.CO1- PO8	MEDIUM	Identify technically and economically feasible problems by working as a team
101003/CS6 22T.CO1- PO9	MEDIUM	Communicate effectively with the engineering community by identifying technically and economically feasible problems
101003/CS6 22T.CO1- PO10	MEDIUM	Demonstrate knowledge and understanding of engineering and management principles by selecting the technically and economically feasible problems
101003/CS6 22T.CO1- PO11	HIGH	Identify technically and economically feasible problems for long term learning.
101003/CS6 22T.CO1- PSO1	MEDIUM	Ability to identify, analyze and design solutions to identify technically and economically feasible problems

101003/CS6 22T.CO1- PSO2	MEDIUM	By designing algorithms and applying standard practices in software project development and Identifying technically and economically feasible problems
101003/CS6 22T.CO1- PSO3	MEDIUM	fundamentals of computer science in competitive research can be applied to Identify technically and economically feasible problems
101003/CS6 22T.CO2- PO1	HIGH	Identify and survey the relevant by applying the knowledge of mathematics, science, engineering fundamentals,
101003/CS6 22T.CO2- PO2	HIGH	Identify, formulate, review research literature, and analyze complex engineering problems get familiarized with software development processes
101003/CS6 22T.CO2- PO3	HIGH	Design solutions for complex engineering problems and design based on the relevant literature
101003/CS6 22T.CO2- PO4	HIGH	Use research-based knowledge including design of experiments based on relevant literature
101003/CS6 22T.CO2- PO5	HIGH	Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes by using modern tools.
101003/CS6 22T.CO2- PO6	MEDIUM	Create, select, and apply appropriate techniques, resources, by identifying and surveying the relevant literature
101003/CS6 22T.CO2- PO7	HIGH	Apply ethical principles and commit to professional ethics based on the relevant literature
101003/CS6 22T.CO2- PO8	MEDIUM	Identify and survey the relevant literature as a team
101003/CS6 22T.CO2- PO9	HIGH	Identify and survey the relevant literature for a good communication to the engineering fraternity.
101003/CS6 22T.CO2- PO10	MEDIUM	Identify and survey the relevant literature to demonstrate knowledge and understanding of engineering and management principles
101003/CS6 22T.CO2- PO11	HIGH	Identify and survey the relevant literature for independent and lifelong learning
101003/CS6 22T.CO2- PSO1	MEDIUM	Design solutions for complex engineering problems by Identifying and survey the relevant literature
101003/CS6	MEDIUM	Identify and survey the relevant literature for acquiring

22T.CO2- PSO2		programming efficiency by designing algorithms and applying standard practices
101003/CS6 22T.CO2- PSO3	MEDIUM	Identify and survey the relevant literature to apply the fundamentals of computer science in competitive research
101003/CS6 22T.CO3- PO1	HIGH	Perform requirement analysis, identify design methodologies by using modern tools & advanced programming techniques and by applying the knowledge of mathematics, science, engineering fundamentals.
101003/CS6 22T.CO3- PO2	HIGH	Identify, formulate, review research literature for requirement analysis, identify design methodologies and develop adaptable & reusable solutions
101003/CS6 22T.CO3- PO3	HIGH	Design solutions for complex engineering problems and perform requirement analysis, identify design methodologies.
101003/CS6 22T.CO3- PO4	HIGH	Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
101003/CS6 22T.CO3- PO5	HIGH	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools
101003/CS6 22T.CO3- PO5	MEDIUM	Perform requirement analysis, identify design methodologies and assess societal, health, safety, legal, and cultural issues.
101003/CS6 22T.CO3- PO6	MEDIUM	Understand the impact of the professional engineering solutions in societal and environmental contexts and Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions
101003/CS6 22T.CO3- PO7	HIGH	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions by applying ethical principles and commit to professional ethics
101003/CS6 22T.CO3- PO8	MEDIUM	Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings
101003/CS6 22T.CO3- PO9	MEDIUM	Communicate effectively with the engineering community and with society at large to perform requirement analysis, identify design methodologies
101003/CS6 22T.CO3- PO10	MEDIUM	Demonstrate knowledge and understanding of engineering requirement analysis by identifying design methodologies
101003/CS6 22T.CO3- PO11	HIGH	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change by analysis, identify design

		methodologies and develop adaptable & reusable solutions.
101003/CS6 22T.CO3- PSO3	MEDIUM	The ability to apply the fundamentals of computer science in competitive research and prior to that perform requirement analysis, identify design methodologies
101003/CS6 22T.CO4- PO1	MEDIUM	Prepare technical report and deliver presentation by applying the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
101003/CS6 22T.CO4- PO2	HIGH	Identify, formulate, review research literature, and analyze complex engineering problems by preparing technical report and deliver presentation.
101003/CS6 22T.CO4- PO3	MEDIUM	Prepare Design solutions for complex engineering problems and create technical report and deliver presentation
101003/CS6 22T.CO4- PO4	MEDIUM	Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions and prepare technical report and deliver presentation
101003/CS6 22T.CO4- PO5	MEDIUM	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools and Prepare technical report and deliver presentation
101003/CS6 22T.CO4- PO7	HIGH	Prepare technical report and deliver presentation by applying ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice
101003/CS6 22T.CO4- PO8	HIGH	Prepare technical report and deliver presentation effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings
101003/CS6 22T.CO4- PO9	HIGH	Communicate effectively with the engineering community and with society at large by prepare technical report and deliver presentation.
101003/CS6 22T.CO4- PO10	MEDIUM	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work by prepare technical report and deliver presentation.
101003/CS6 22T.CO4- PO11	HIGH	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change by prepare technical report and deliver presentation
101003/CS6 22T.CO4- PSO1	MEDIUM	Prepare a technical report and deliver presentation to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas.
101003/CS6	MEDIUM	To acquire programming efficiency by designing algorithms

22T.CO4-PSO2		and applying standard practices in software project development and to prepare technical report and deliver presentation
101003/CS6 22T.CO4-PSO3	MEDIUM	To apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs by preparing technical report and deliver presentation
101003/CS6 22T.CO5-PO1	HIGH	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
101003/CS6 22T.CO5-PO2	HIGH	Identify, formulate, review research literature, and analyze complex engineering problems by applying engineering and management principles to achieve the goal of the project.
101003/CS6 22T.CO5-PO3	HIGH	Apply engineering and management principles to achieve the goal of the project and to design solutions for complex engineering problems and design system components or processes that meet the specified needs
101003/CS6 22T.CO5-PO4	MEDIUM	Apply engineering and management principles to achieve the goal of the project and use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
101003/CS6 22T.CO5-PO5	MEDIUM	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools and to apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5-PO6	MEDIUM	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities by applying engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5-PO7	MEDIUM	Understand the impact of the professional engineering solutions in societal and environmental contexts, and apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5-PO8	HIGH	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice and to use the engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5-PO9	MEDIUM	Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings and to apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5-	MEDIUM	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as

PO11		a member and leader in a team. Manage projects in multidisciplinary environments and to apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5- PO11	HIGH	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change and to apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5- PSO1	MEDIUM	The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas Apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5- PSO2	MEDIUM	The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry and to apply engineering and management principles to achieve the goal of the project
101003/CS6 22T.CO5- PSO3	MEDIUM	The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur and apply engineering and management principles to achieve the goal of the project